



Beyond planktonic encounter theory: a Rotimer-based model of benthic mate search in the rotifer *Euchlanis dilatata*

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Abstract The benthic rotifer *Euchlanis dilatata* produces Rotimer, a filamentous biopolymer forming persistent three-dimensional webs on substrate surfaces. Rotimer is conserved across Rotifera and serves basal functions in females, but its function in non-feeding, short-lived males remains unclear. We test whether female-produced Rotimer functions as a mate-search cue detectable by males. We propose the Benthic Rotifer Encounter Model (BREM), extending planktonic encounter theory to substrate-anchored systems by substituting the Rotimer web's effective encounter radius for the female body radius; BREM

is bidirectional in principle, though only the female-to-male direction was tested here. Consistent with BREM, female webs were structurally more complex than male webs, and males showed greater trajectory straightness in the presence of female Rotimer. Directed tracking of conspecific Rotimer to copulation was observed in both sexes. These findings support BREM's female–male encounter pathway and generate testable predictions for mate-search dynamics in benthic microinvertebrates.

Keywords *Euchlanis dilatata* · Rotimer · Benthic mate search · Encounter-rate model · Biopolymer web · Monogononta

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Introduction

Mate search in microscopic invertebrates presents a fundamental challenge: individuals must locate conspecific mates across complex microenvironments at densities where random encounter is often insufficient to sustain sexual reproduction (Gerritsen & Strickler, 1977; Snell & Garman, 1986). The probability of male–female contact depends on the movement speeds of both sexes, the duration of male fertility, and the effective encounter distance between them. For planktonic rotifers such as *Brachionus plicatilis* Müller 1786, these parameters are formalized in the Gerritsen-Strickler encounter-rate model (Eq. 1), in which the encounter rate Z

depends on the female body radius d , female density N , and the velocities of both sexes (Gerritsen & Strickler, 1977; Snell & Garman, 1986). In *Brachionus*, mate recognition by males depends on contact chemoreception of a 29 kDa glycoprotein on the female body surface (Snell et al., 1995; Gómez & Serra, 1996; Snell & Stelzer, 2005), so the effective encounter distance d is constrained to approximately the female body radius, and in this system females play no active role in mate location (Gilbert, 1963; Snell & Hawkinson, 1983; Rico-Martínez & Snell, 1997; Gurarie & Ovaskainen, 2013). Snell & Garman (1986) showed that under these conditions encounter probability approaches one only above approximately 1,000 females per liter.

Equation 1. Planktonic encounter rate (Gerritsen & Strickler, 1977; Snell & Garman, 1986)

$$Z = \frac{\pi d^2 N}{3} \cdot \frac{v_m^2 + 3v_f^2}{v_m} \quad (1)$$

d = encounter distance (female body radius, μm), N = female density (individuals per unit volume), v_m = male velocity ($\mu\text{m s}^{-1}$); v_f = female velocity ($\mu\text{m s}^{-1}$).

Valid for $v_m \geq v_f$; in contact chemoreception systems d equals the female body radius.

Benthic rotifers such as *Euchlanis dilatata* Ehrenberg, 1830, a cosmopolitan species that encompasses cryptic lineages with reproductive isolation (Kordbacheh et al., 2017, 2019), inhabit a fundamentally different physical environment. They lead a periphytic lifestyle on substrate surfaces, deposit eggs directly onto the substrate (Li et al., 2019), and produce the Rotimer: a viscoelastic, proteinaceous exogenic biopolymer that forms a persistent three-dimensional web anchored to substrate attachment points, with individual filaments reaching up to 1.3 mm above the substrate surface into the overlying water column (Balazs et al., 2021). Rotimer production is broadly conserved across the phylum Rotifera, including asexual bdelloid rotifers (Segers, 2007; Datki et al., 2021). In *E. dilatata* females, Rotimer serves multiple basal functions: particle-dependent mechanical stimulation triggers Rotimer secretion and egg laying, after which Rotimer fixes eggs to the substrate and organizes the local feeding microenvironment (Datki et al., 2021, 2022). Substrate organization by female Rotimer may additionally structure the physical space that searching males navigate, a possibility consistent

with BREM predictions and discussed further in the Implications section.

Despite producing Rotimer, *E. dilatata* males lack the life-history context required for its known basal functions: they are non-feeding with a degenerate digestive system and do not lay eggs, precluding benefits from substrate organization, egg fixation, or feeding enhancement. Males remain actively motile and fertile for only ~48 h, after which locomotor capacity declines markedly prior to death (Z. Datki, pers. comm.; B.J. Dingmann, unpubl. obs.; Snell & Childress, 1987; Kordbacheh et al., 2019). Here we test the hypothesis that female-produced Rotimer functions as a mate-search cue detectable by males, providing the empirical basis for the present study; the broader question of male Rotimer's own reproductive function is taken up in the Discussion.

Aquatic invertebrates have evolved diverse mechanisms for extending the mate-search signal radius beyond body contact, including transient hydrodynamic wake trails in copepods (Snell & Morris, 1993; Yen & Lasley, 2010) and substrate-anchored silk draglines carrying contact pheromones in cursorial spiders (Beyer et al., 2023). Benthic mate-search systems mediated by dissolved chemical gradients have been described in the diatom *Seminavis robusta* Danielidis & Mann, 2002 (Bondoc et al., 2016), but these model chemotaxis toward gradients in the water column above the substrate rather than encounter with a persistent solid structure. None of these frameworks address encounter with a persistent, substrate-anchored three-dimensional biopolymer.

Here we introduce the Benthic Rotifer Encounter Model (BREM), a theoretical framework that extends the Gerritsen-Strickler encounter-rate model to substrate-anchored biopolymer systems and generates testable predictions for mate-search behavior in benthic microinvertebrates. BREM retains the logical structure of Eq. 1 but replaces the female body radius d with d_R , the effective encounter radius of the Rotimer web volume (Eqs. 2a and b). Because both sexes produce Rotimer, BREM is formulated as a bidirectional model: the female web forms an encounter zone for searching males, and the male web may form an analogous encounter zone for searching females, a possibility developed further in the Discussion. The structural differences between male and female webs documented here raise the additional hypothesis that female web morphology provides

a contact-readable signal enabling conspecific discrimination and directed search by males; whether male web morphology serves an analogous cue function for females was not tested here. By substituting the Rotimer web volume for the female body radius, BREM predicts a substantially expanded encounter radius relative to body-contact distance (quantified in the Discussion), with web persistence allowing encounter probability to accumulate over the male's benthic residence time. This distinction may be ecologically significant for *E. dilatata*, which inhabits structured littoral habitats where local densities of sexually reproducing individuals may be low. Empirical field density estimates for benthic *E. dilatata* populations are not currently available in the literature.

Equation 2 Rotimer-mediated encounter rate and contact probability (male searching female-produced Rotimer) (BREM)

$$Z_R = \frac{\pi d_R^2 N}{3} \cdot \frac{v_m^2 + 3v_f^2}{v_m} \quad (2a)$$

$$P_R = 1 - e^{-Z_R t_m} \quad (2b)$$

d_R = effective encounter radius of female Rotimer web volume (μm), t_m = duration of male fertility (s); P_R = Rotimer contact probability

Bounds on d_R : lower $\approx 50 \mu\text{m}$ (mean female radius); upper $\approx 1,300 \mu\text{m}$ (Datki et al., 2021).

Encounter ratio: $Z_R/Z = (d_R/dbody)^2 = 1$ to 676 for d_R ranging from body radius ($dbody \approx 50 \mu\text{m}$; ratio = 1, baseline) to $d_R \approx 1,300 \mu\text{m}$ (Datki et al., 2021; ratio ≈ 676). Values below 1 are not biologically meaningful; the model applies only when d_R exceeds body-contact distance. Thus, encounter rate scales with the square of the effective encounter radius, making even modest increases in d_R disproportionately impactful.

Equation 2c. Mirrored encounter rate for the reverse search direction (female searching male-produced Rotimer)

$$Z_R \varphi = \frac{\pi d_R^2 \sigma N_m}{3} \cdot \frac{v_m^2 + 3v_f^2}{v_m} \quad (2c)$$

$d_R \varphi$ = effective encounter radius of male Rotimer web volume (μm); N_m = male density (individuals

per unit volume). The velocity term retains the same structure as Eq. 2a, because relative encounter velocity depends on which sex moves faster in absolute terms, not on which sex's web is being detected. Although male velocity in body lengths per second exceeds female velocity approximately 1.8-fold due to smaller male body size (male: 5.7 body lengths/s; female: 3.2 body lengths/s), the encounter equations use absolute velocities because both the Rotimer encounter zone and male fertility duration are measured in absolute spatial and temporal units. This parameterization ensures that Eqs. 2a and c share identical mathematical structure when $v_m = v_f$.

Valid for $v_m \geq v_f$ in contact chemoreception systems (Gerritsen & Strickler, 1977; Snell & Gorman, 1986); however, in *E. dilatata* $v_m = v_f$ in absolute terms (Rico-Martínez & Snell, 1997). Female velocity was not directly measured in the present study, so Eq. 2c cannot yet be numerically parameterized with our data and is presented to complete the bidirectional formulation of BREM introduced in Eqs. 2a–b.

The present study provides preliminary empirical tests of three predictions generated by BREM: 1. If both sexes' Rotimer webs function as encounter zones, male and female webs should be structurally distinguishable, providing a physical basis for sex-specific detection. 2. If female Rotimer constitutes an encounter zone for searching males, males should exhibit more directed locomotor behavior, movement through, rather than random exploration of, a female encounter zone, when female Rotimer is present, compared with males in the absence of female Rotimer consistent with increased contact probability. 3. If conspecific Rotimer functions as a directional cue, individuals should exhibit directed movement along Rotimer strands toward the producing individual; because the present design does not permit isolation of Rotimer from living individuals, this prediction is evaluated through observational documentation only. These predictions are tested through skeletal analysis of sex-specific Rotimer web architecture, behavioral tracking of males with and without female Rotimer, and observational documentation of tracking and copulation events. The present study is not intended to parameterize BREM fully, but to provide initial empirical constraints on key assumptions required for its future quantitative evaluation.

Materials and methods

Study organism and collection

Euchlanis dilatata were collected from Blood Lake (GPS coordinates (WGS84 datum): 46° 16' 25" N; 20° 08' 39" E; September and October), Szeged, Southern Great Plain, Hungary, the same source population described in Datki et al. (2021, 2022). Animals were isolated using a micropipette under an inverted light microscope and identified to species based on morphological keys (Koste, 1978). Males and females were distinguished by morphology; males of *E. dilatata* are recognizable by their reduced body size and degenerate digestive system. Animals were maintained in standard medium during experiments (mg/L: Ca^{2+} 31.05; Mg^{2+} 17.6; Na^+ 0.9; K^+ 0.25; Fe^{2+} 0.001; HCO_3^- 153.097; SO_4^{2-} 3; Cl^- 0.8; F^- 0.02; H_2SiO_3 3.3; pH=7.5) at 22°C, 40% air humidity (Datki et al., 2021).

Rotimer web collection and structural analysis

Male- and female-produced Rotimer webs were prepared separately following the Rotimer-Inductor Conglomerate (RIC) induction protocol of Datki et al. (2021). Two individuals of the same sex (males or females; $n=2$ per well, one sex per well) were placed in 50 μL of standard medium in individual wells of 24-well plates with epoxy metal beads (Dynabeads M-270; Life Technologies AS; cat. no. 14301; $\sim 2.8 \mu\text{m}$; 50 $\mu\text{g/mL}$) for 20–30 min (22 °C, pH 7.5, under standard laboratory fluorescent lighting, 12:12 h light:dark photoperiod). A total of 10 wells per sex were used, each constituting an independent Rotimer preparation. Animals were removed by micropipette; preparations were dried at 22 °C, 40% humidity for 30 min and imaged by using a Leica DMS300 digital microscope system with integrated camera at 2.5 $\times$ objective magnification (the same system and magnification used for behavioral tracking; see Male behavioral tracking experiment). Images were converted to binary format (1 $\mu\text{m}=2.46$ pixels; 8-bit) and analyzed using the Analyze Skeleton plugin in Fiji/ImageJ (Arganda-Carreras et al., 2010). Structural metrics are reported in both μm (converted values) and pixels (raw instrument units) to facilitate direct comparison with future studies using different calibration values; metrics included

branch length, Euclidean distance between endpoints, running average branch length, branch count, junction count, triple and quadruple branch points, slab voxels, and end-point voxels ($n=10$ webs per sex). There is no evidence that sexual and asexual females produce structurally different Rotimer; observed differences are interpreted as sex-based rather than morph-based. Structural differences were assessed using Welch two-sample t-tests; means are reported $\pm$ standard deviation.

Male behavioral tracking experiment

Two treatments were established. In the male-only treatment, two males were placed in 50 μL of standard medium (no females, no prior Rotimer deposition). In the mix (male+female Rotimer) treatment, two males were placed in 50 μL of standard medium together with four females; females were introduced 15 min before the start of tracking to allow natural Rotimer deposition onto the substrate prior to observation, and remained present during tracking epoxy metal beads; as described above were added to both treatments to visualize Rotimer as a Rotimer-Inductor Conglomerate (RIC); additional observations without beads confirmed behavioral consistency between bead-present and bead-absent conditions. Tracking of males began after the same elapsed time in both treatments. We acknowledge that the two treatments differed in total individual count (2 vs. 6 organisms) and that the pre-deposition period was available only in the mix treatment; these design asymmetries are discussed as limitations below. Sexual and asexual female morphs are morphologically identical in *E. dilatata* and cannot be distinguished visually; female Rotimer preparations therefore represented a mixed-morph population. A total of 11 males were tracked in the male-only treatment (56 tracks, 6 videos) and 10 males in the mix treatment (39 tracks, 7 videos).

Male movement was recorded using a Leica DMS300 digital microscope system with integrated camera at 2.5 $\times$ objective magnification (spatial calibration: 0.1260 pixels/ μm) or 3.2 $\times$ objective magnification (0.1600 pixels/ μm ; applied to videos 7 and 9 only) at 30 frames per second. Males were tracked frame-by-frame using MTrackJ (Meijering et al., 2012) in Fiji/ImageJ (Schindelin et al., 2012) with every fifth frame recorded as a point (equivalent to one point per 0.167 s); recording every fifth

frame rather than every frame was necessary because single-frame displacements at 30 fps were too small to resolve reliably given the organisms' movement speed, and this sampling interval was applied consistently across all tracks.

Movement metric calculation

MTrackJ export files were analyzed in R (R Core Team, 2024) using the tidyverse and ggpvr packages. Step-level metrics were calculated for each consecutive frame pair, including step displacement (μm), instantaneous velocity ($\mu\text{m s}^{-1}$), step heading (degrees, defined as the angle of each displacement vector relative to the positive x-axis of the coordinate frame established at track origin), and turning angle (degrees, normalized to -180° to $+180^\circ$, where positive values indicate right turns). Track-level metrics were then calculated for each tracked segment, including mean velocity, total distance traveled (sum of step displacements), net displacement (distance between the first and final coordinates of the track), trajectory straightness (net displacement / total distance traveled), and mean absolute turning angle.

To treat each male as an independent observational unit, track-level metrics were averaged across all tracks for a given male, yielding a single mean per male; these male-level means ($n=11$ male-only; $n=10$ mix) were the unit of analysis for all subsequent statistical comparisons. This male-level-mean approach was used in preference to a mixed-effects model with male identity as a random effect because averaging track-level metrics to a single value per male already makes the male, the true biological replicate in this design, the unit of analysis; with a modest and unbalanced number of tracks per male (1–6 tracks per animal across videos), a mixed model would need to estimate a male-level variance component from very limited within-male replication, risking unstable variance estimates at this sample size. Tracking duration targeted two minutes per male but varied due to loss of the male from the field of view or inability to initiate tracking within the first two minutes of each video; mean tracking duration was 148 ± 107 s in the male-only treatment and 88 ± 22 s in the mix treatment. The high standard deviation in the male-only treatment reflects variation in male residence time within the field of view across videos. Total distance was therefore normalized by each

male's own recorded tracking duration (i.e., distance / that male's recorded duration, then rescaled $\times 120$ s), so that variation in observation time among males is accounted for before values are expressed on a common 120 s timescale for comparison across males.

Treatment effects were tested using linear models with male-level means as the response variable (metric ~ treatment). Model assumptions were evaluated using residual diagnostics, and Wilcoxon rank-sum tests were applied as non-parametric robustness checks. Effect sizes for treatment comparisons were calculated as Cohen's $d = (M_{\text{mix}} - M_{\text{male-only}}) / SD_{\text{pooled}}$, where $SD_{\text{pooled}} = \sqrt{[(SD_1^2 + SD_2^2) / 2]}$; post-hoc observed power and the sample size required for 80% power at the observed effect size were calculated using the pwr package in R (Champely, 2020), $\alpha=0.05$, two-sided (see Table S1). All analyses and figures were conducted in R (R Core Team, 2024), with analysis scripts and processed datasets provided in the Supplementary Material to enable full reproduction of results.

Use of AI assistance

Claude (Anthropic) was used to assist in manuscript drafting/editing, structural organization, and mathematical framework development during preparation of this manuscript. All scientific content, data, interpretations, and final text were approved by the authors, who take full responsibility for the accuracy and integrity of the work.

Results

Structural differences between male and female Rotimer webs

Female Rotimer webs were significantly more complex than male webs across nearly all measured parameters (Fig. 1). Mean branch length was significantly greater in female webs ($1.76 \pm 0.37 \mu\text{m}$; 4.32 ± 0.90 px) than male webs ($1.39 \pm 0.32 \mu\text{m}$; 3.41 ± 0.78 px; $t = -2.43$, $df = 17.7$, $P = 0.026$), as was total branch length (female: $14,310 \pm 7,076 \mu\text{m}$; $35,202 \pm 17,409$ px; male: $7,655 \pm 4,158 \mu\text{m}$; $18,831 \pm 10,229$ px; $t = -2.56$, $df = 14.6$, $P = 0.022$), mean Euclidean distance between branch endpoints (female: $1.48 \pm 0.25 \mu\text{m}$; 3.64 ± 0.61 px; male:

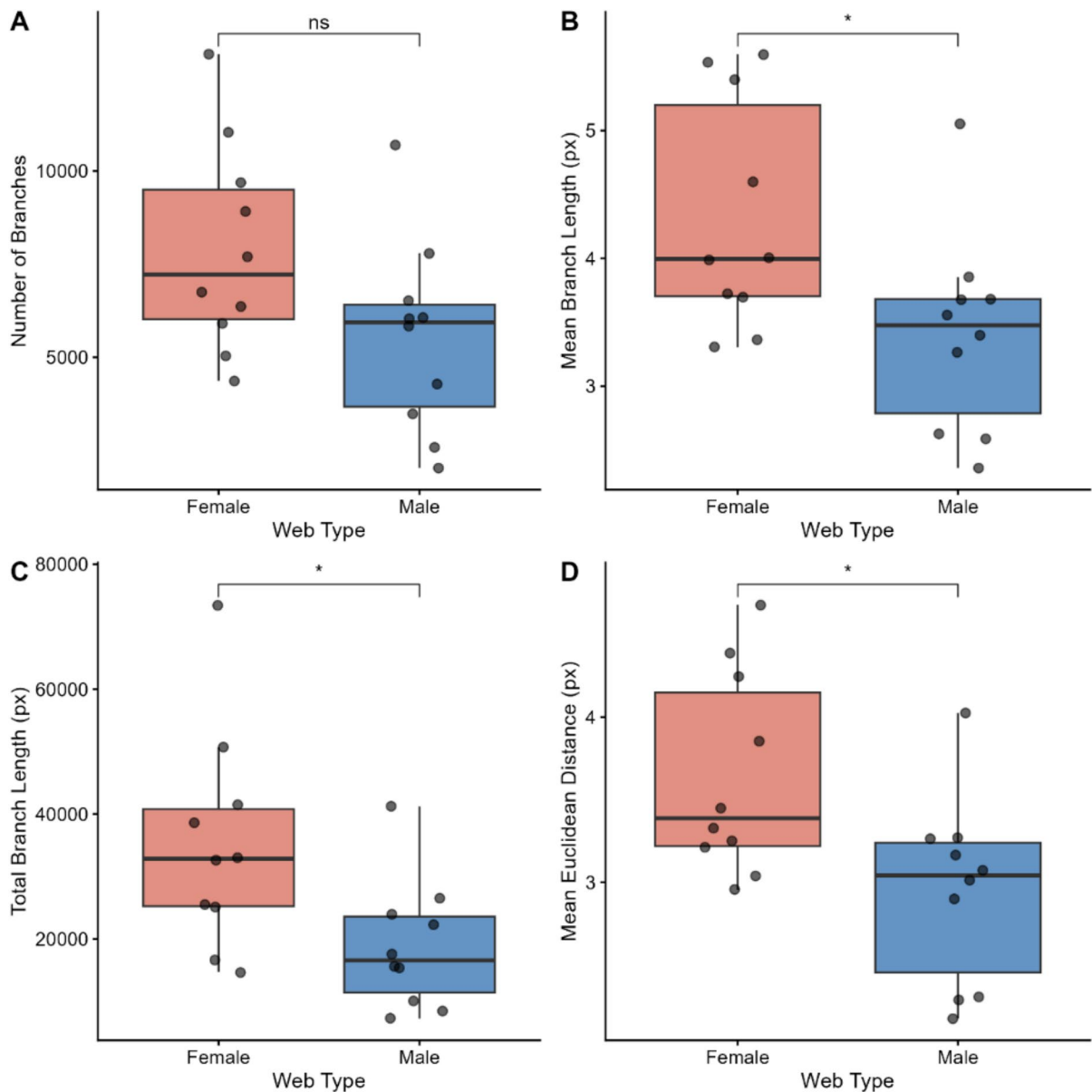


Fig. 1 Structural comparison of male and female Rotimer webs. Boxplots show (A) number of branches, (B) mean branch length (px), (C) total branch length (px), and (D) mean Euclidean distance between branch endpoints (px). Individual data points are overlaid. Female webs exhibited significantly greater structural complexity than male webs across

most parameters (see Results for full statistics); the number of branches (A) did not reach significance. $n=10$ web preparations per sex. The full set of significant structural metrics beyond these four representative panels is reported in the Results and Supplementary Table S2. Boxes show interquartile range; horizontal line = median; whiskers extend to $1.5 \times \text{IQR}$

$1.20 \pm 0.23 \mu\text{m}$; $2.95 \pm 0.57 \text{ px}$; $t = -2.63$, $\text{df} = 17.9$, $P = 0.017$), and mean running average branch length (female: $1.43 \pm 0.36 \mu\text{m}$; male: $1.19 \pm 0.18 \mu\text{m}$; $t = -2.61$, $\text{df} = 17.8$, $P = 0.016$). Female webs also had significantly greater branch count ($P = 0.013$),

junction voxels ($P = 0.043$), triple points ($P = 0.025$), quadruple points ($P = 0.027$), slab voxels ($P = 0.014$), end-point voxels ($P = 0.008$), average branch length ($P = 0.010$), and maximum branch length ($P = 0.012$); complete t - and df -values for all comparisons are

provided in Table S2. Number of branches did not reach significance ($P=0.065$).

Male locomotor behavior in the presence of female Rotimer

Time-normalized total distance (male-only: $75,925 \pm 30,550$ μm per 120 s; mix: $57,085 \pm 18,839$ μm per 120 s) did not differ significantly between treatments ($F=1.99$, $P=0.174$; Wilcoxon $W=78$, $P=0.115$; data not shown). Likewise, mean velocity (male-only: 636 ± 317 $\mu\text{m/s}$; mix: 478 ± 154 $\mu\text{m/s}$; $P=0.168$) and mean turning angle (male-only: $33.5 \pm 31.4^\circ$; mix: $40.4 \pm 20.6^\circ$; $P=0.564$) did not differ significantly between treatments. However, males in the presence of female Rotimer showed significantly higher trajectory straightness (male-only: 0.256 ± 0.122 ; mix: 0.438 ± 0.164 ; $F_{1,19}=8.39$, $P=0.009$; Wilcoxon $W=20$, $P=0.015$; Fig. 2), indicating more directed, less tortuous paths when female

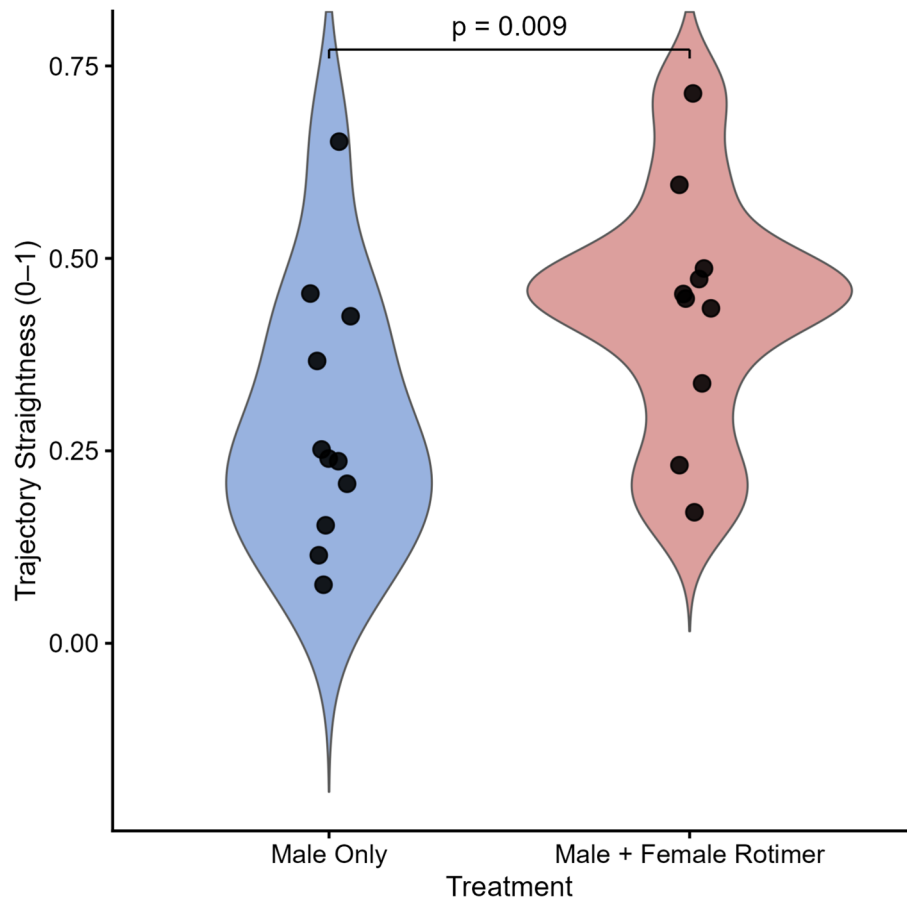
Rotimer was present. Males in the male-only treatment typically produced looping, spatially confined paths, whereas males in the mix treatment produced more open, directional trajectories. Representative movement tracks illustrating this contrast in path geometry are shown in Fig. 3.

Modeled encounter-rate (Z_R) and contact-probability (P_R) values under BREM (Eqs. 2a–b), computed from the velocity, male fertility duration, and density parameters detailed in the Discussion section, are summarized in Table 1; their ecological interpretation is developed there.

Discussion

These findings link Rotimer web structure to sex-specific movement behavior and provide initial empirical support for BREM. Female Rotimer webs were structurally more complex than male webs, consistent with

Fig. 2 Trajectory straightness of male *Euchlanis dilatata* in the male-only and male + female Rotimer treatments, testing Prediction 2 of the Benthic Rotifer Encounter Model (BREM). Violin plots show the distribution of individual male-level straightness index values (0–1); jittered points represent individual males (each derived from means across all tracks for that male). Males in the presence of female Rotimer showed significantly higher trajectory straightness than males in the male-only treatment (see Results for full statistics), consistent with more directed movement within a female Rotimer encounter zone. $n=11$ males in the male-only treatment; $n=10$ males in the male + female Rotimer treatment



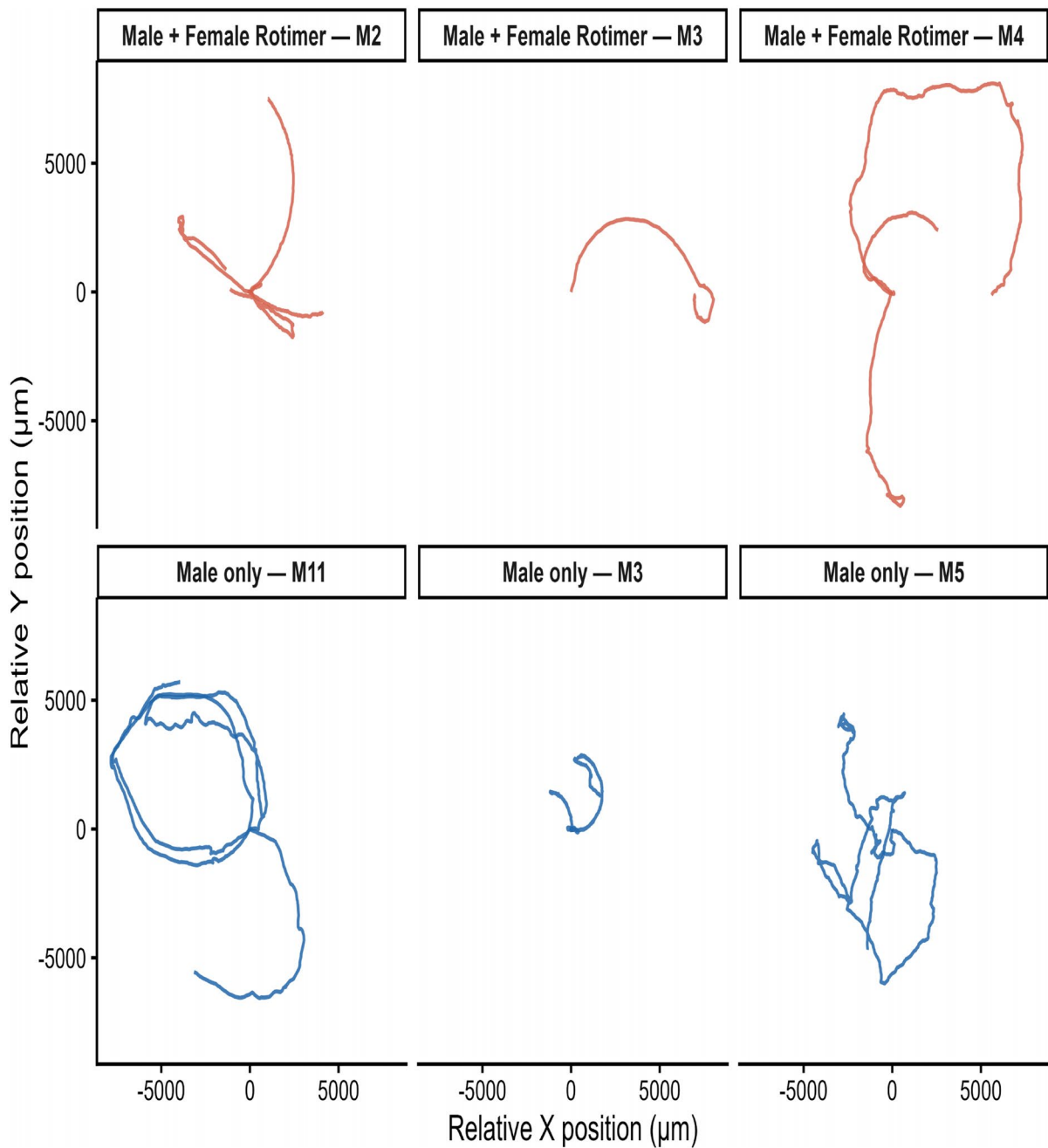


Fig. 3 Representative movement tracks of male *Euchlanis dilatata* in the male+female Rotimer (top row, coral) and male-only (bottom row, blue) treatments. Tracks are normalized to a common origin (0, 0). Path-geometry differences between treatments are described in the Results. Representa-

tive tracks (rather than overlay of all tracks) are shown to illustrate qualitative differences in path shape (see Fig. 2 for the quantitative population-level variability in path geometry). Individual male identifiers (M2–M5, M11) are shown in panel titles. Axes in micrometers (μm)

Table 1 Modeled encounter rate (Z_R) and contact probability (P_R) under the Benthic Rotifer Encounter Model (BREM; Eqs. 2a–b) for *Euchlanis dilatata*, across the empirically bounded range of effective encounter radius (d_R).

d_R (μm)	Basis	Z_R (enc s^{-1})	Z_R fold vs. baseline	P_R ($t_m = 48$ h, $N = 0.5$ mL^{-1})	P ($t_m = 48$ h, $N = 0.001$ mL^{-1})
50	Body contact (baseline)	5.13×10^{-3}	1 $\times$ (baseline)	≈ 1.000	0.83
1,300	Max web extent (Datki et al., 2021)	347	67,600 $\times$	≈ 1.000	≈ 1.000

Full justification for parameters and justification (velocity, male fertility duration, and density assumptions) are given in the Results. All encounter radius estimates are conservative lower bounds relative to native three-dimensional web geometry

Prediction 1, and males exhibited significantly more directed movement in the presence of female Rotimer, consistent with Prediction 2. Directed tracking of conspecific Rotimer to copulation was observed in both sexes, consistent with Prediction 3. Together, these results suggest Rotimer may function as a spatially extended encounter medium, though key uncertainties regarding detection mechanisms and three-dimensional web geometry remain and are addressed in turn below.

Male and female Rotimer webs differed markedly in structural complexity. Consistent with BREM predictions, female Rotimer webs were more structurally complex than male webs, including greater total length, branching, and junction density. These differences likely reflect the multiple functional demands placed on female Rotimer, including feeding micro-environment construction, substrate organization, and egg fixation.

By contrast, male webs were less complex, consistent with a more specialized role in signal deposition rather than long-term substrate modification. Note that all structural measurements were obtained from desiccated two-dimensional preparations and likely underestimate divergence in native three-dimensional webs (see Limitations). If Rotimer structure influences encounter dynamics, such differences are expected to translate into sex-specific movement responses.

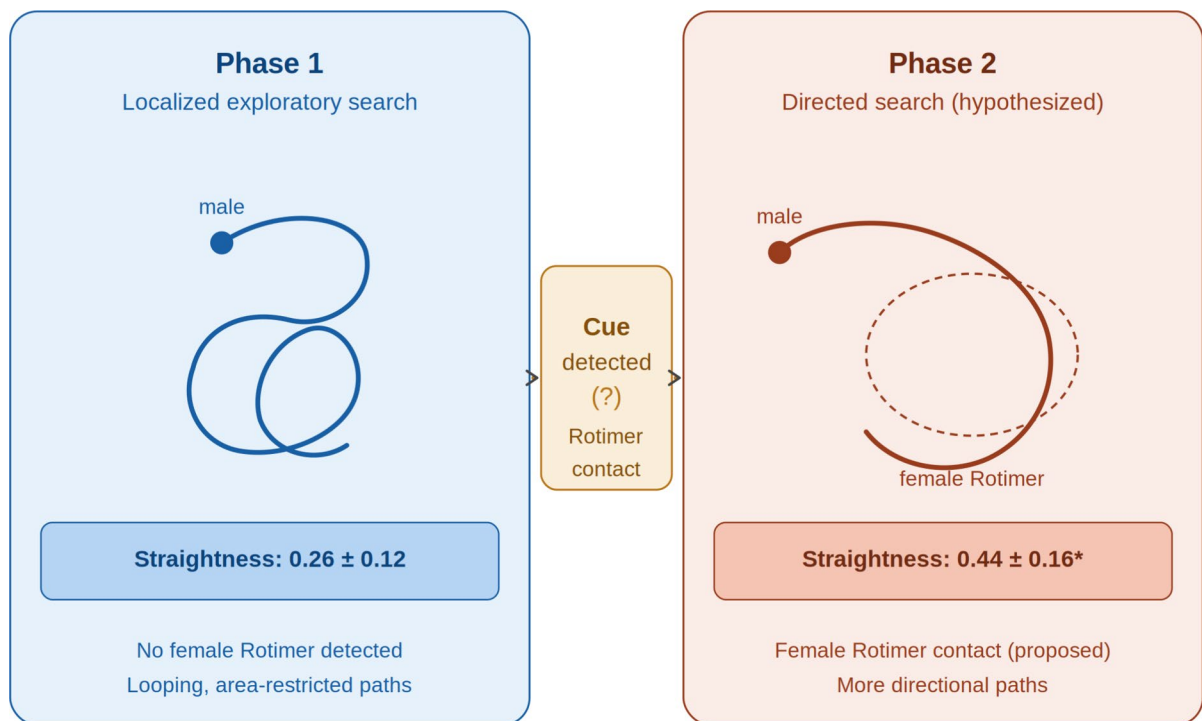
Male search behavior also responded to female Rotimer. Males showed significantly greater trajectory straightness in the presence of female Rotimer, indicating more directed movement within a female-derived encounter zone.

An important interpretive constraint is that the mix treatment included living females alongside deposited

Rotimer. Males could therefore respond to female-derived waterborne cues, direct contact, or Rotimer itself. However, mate recognition in rotifers is not known to involve long-range diffusible chemical signaling and is instead mediated by contact chemoreception. Within this framework, any Rotimer-based signaling mechanism is most plausibly contact-dependent (Snell et al., 1995; Snell & Stelzer, 2005).

Although the present design cannot isolate these mechanisms, the behavioral response is robust, and its sensory basis represents a clear target for future work. These behavioral phases resemble area-restricted search, a well-documented strategy in which organisms shift from localized, tortuous movement to directed, extensive movement upon detecting a resource-associated cue (Dorfman et al., 2022) and are consistent with a hypothesized two-phase mate-search strategy illustrated in Fig. 4, proposed here as a BREM-derived hypothesis for future experimental testing.

These results indicate that males adjust their search patterns within female-derived encounter zones. A comparable transition from exploratory to directed movement upon trail detection has been documented kinematically in mate-tracking copepods (Weissburg et al., 1998), lending independent plausibility to a similar mechanism in *E. dilatata*. Contact-mediated detection of Rotimer is a plausible mechanism given that mate recognition in rotifers is generally mediated by contact chemoreception rather than long-range diffusible signals (Snell et al., 1995; Snell & Stelzer, 2005), but whether structural, chemical, or mechanosensory properties of Rotimer filaments drive the observed directed movement remains to be determined. Although these behavioral patterns align with BREM predictions, they do not by themselves



* $p = 0.009$; trigger mechanism unknown — hypothesized framework for future testing

Fig. 4 Hypothesized two-phase male mate-search behavior predicted by the Benthic Rotifer Encounter Model (BREM). Phase 1 (male-only treatment; Trajectory straightness = net displacement / total distance (see Methods)). Phase 1 (male-only treatment; mean straightness 0.26 ± 0.12) represents localized exploratory movement in the absence of female Rotimer. Phase 2 represents the hypothesized transition to more directed

movement following female Rotimer contact (see Results for statistics). The cue-detection trigger (center panel; marked with ?) is attributed to Rotimer contact based on the experimental design, but the specific sensory mechanism remains unknown. This schematic is a BREM-derived hypothesis for future experimental testing and does not constitute a demonstrated mechanistic model

establish the underlying sensory mechanism, which may involve chemical, mechanical, or structural cues associated with Rotimer.

The parsimony of a reproductive role for male Rotimer is considered next. Rotimer production is broadly conserved across rotifer lineages, including bdelloids, where it serves basal ecological functions such as substrate organization, feeding microenvironment construction, and egg fixation (Datki et al., 2021). In *E. dilatata*, these functions are fulfilled by females, whose webs are structurally more complex and consistent with multiple concurrent roles. Given that males are non-feeding, short-lived, and do not lay eggs, a reproductive or mate-search function provides the most parsimonious explanation for male Rotimer production.

Preliminary behavioral observations are consistent with this interpretation. Directed tracking of conspecific Rotimer to copulation was observed in both sexes, including a single instance in which a female followed a male's trajectory to contact and mating (see Data availability). Together with the male-directed tracking documented above, this raises the possibility that females may also track male-deposited Rotimer; however, this interpretation rests on very limited footage (a single video), and female behavior has not been systematically compared across conditions (e.g., presence of other females vs. other males) to rule out alternative explanations. We therefore present this observation as anecdotal, motivating future study, rather than as an adequately tested finding or one that meaningfully supports the model's

conclusions; direct experimental validation will be required to confirm this mechanism.

These findings carry several implications for BREM. BREM describes a system in which both sexes contribute to encounter dynamics through Rotimer production, but not equally. Female web geometry suggests a larger effective encounter radius ($d_{R,\text{♀}}$) relative to male webs ($d_{R,\text{♂}}$), and therefore a disproportionate contribution to overall encounter probability—an asymmetry consistent with reproductive roles while maintaining bidirectional signaling. This architecture parallels other substrate-anchored mate-search systems: in the diatom *Seminavis robusta*, pheromones deposited on the substrate surface guide partner approach (Bondoc et al., 2016), and in cursorial spiders, silk draglines carrying contact pheromones direct male search toward females (Beyer et al., 2023). BREM extends this class of substrate-mediated encounter frameworks to three-dimensional persistent biopolymer webs, where the volumetric extent and structural complexity of the web, rather than chemical gradient diffusion, define the effective encounter radius. The bidirectional nature of BREM also distinguishes it from planktonic encounter theory: whereas encounter in *Brachionus* depends entirely on passive female presence, both *E. dilatata* sexes contribute to the encounter space through web deposition, consistent with the bidirectional tracking observed here.

Importantly, BREM is intended as a hypothesis-generating framework: while the patterns reported here are consistent with Rotimer-mediated encounter processes, experimental manipulation of web presence, structure, and detection pathways will be required to parameterize and test the model quantitatively.

Several limitations should be noted. First, as discussed in the Behavioral responses section, the mix treatment contained living females alongside deposited Rotimer and the two treatments were not fully parallel in organism count or pre-deposition time; the observed behavioral response therefore cannot be attributed exclusively to Rotimer-mediated detection. A future design in which a single male is introduced to a well previously conditioned by female Rotimer (females removed) versus an unconditioned well would permit stronger causal inference and is a clear priority. A complementary design would compare male behavior in the presence of a live female

in fresh medium without prior Rotimer deposition against male behavior in the presence of a live female with deposited Rotimer, which would help distinguish the effect of female presence itself from the effect of deposited Rotimer. Discriminating among candidate mechanisms, contact chemoreception, mechanosensory detection of filament structure, or combined chemo-mechanical cues, will require experiments that isolate deposited Rotimer from living females while preserving native web architecture, a technically challenging task given the fragile and viscoelastic nature of the material. Second, sample sizes were modest ($n=11$ and $n=10$), and results should be considered preliminary; however, the large effect size for trajectory straightness (Cohen's $d \approx 1.26$) suggests the primary finding is likely to replicate, whereas non-significant metrics were underpowered to detect small-to-moderate effects and should not be interpreted as evidence of null effects. Third, structural measurements were derived from desiccated two-dimensional preparations and likely underestimate divergence in native three-dimensional webs, so encounter radius estimates should be considered conservative. Finally, all animals were drawn from a single population, limiting inference across cryptic lineages. Additionally, Rotimer web preparations pooled two individuals per well, so structural metrics reflect group-level web output rather than individually attributable replicates; single-individual wells would allow future studies to treat web measurements as independent replicates at the individual level. Together, these limitations define clear priorities for future experimental and quantitative investigation.

The velocity and density parameters used in BREM (Eqs. 2a–b) are drawn from published values for *E. dilatata*. *Velocity*: described in the present study ranged from 478–636 $\mu\text{m/s}$ depending on treatment, consistent with values reported by Datki et al. (2021; 465 $\mu\text{m/s}$); use of the higher published estimate yields conservative encounter-rate calculations. Male fertility duration: $t_m = 172,800$ s (48 h), based on field observations of active male motility and fertility (see Introduction); short male fertility windows are documented in other monogononts (Snell & Childress, 1987); this duration has not been formally measured in *E. dilatata* and the estimate reflects the active search phase, excluding the terminal locomotor decline observed prior to death. Density: $N=0.5$ ind mL^{-1} is an illustrative value consistent with

experimental starting densities used in published laboratory studies of *E. dilatata* (Walsh, 1995; Nandini et al., 2011); empirical field density estimates for benthic *E. dilatata* populations are not currently available. The low-density comparison ($N=0.001$ ind mL^{-1}) illustrates model behavior under ecologically sparse conditions, where differences in P_R become meaningful. Table 1 summarizes the resulting modeled Z_R and P_R values across the empirically bounded range of d_R (see Table 1 caption for a brief summary; full parameter justification is above). The encounter rate under maximum web geometry ($d_R=1,300 \mu\text{m}$) is approximately 67,600-fold greater than under body-contact detection, and approximately 146-fold greater under the conservative lower-bound web radius from this study. At the illustrative laboratory density ($N=0.5$ ind mL^{-1}), P approaches 1.0 under all d scenarios because the 48-h male fertility window provides ample encounter time even at body-contact distance. The low-density comparison ($N=0.001$ ind mL^{-1}) illustrates model behavior under ecologically sparse conditions: body-contact detection yielding $P_R=0.83$ compared to near-certainty under maximum web extent, so P_R differences become ecologically meaningful at low density. These calculations underscore the potential significance of Rotimer-mediated encounters for a benthic species at low population density.

Future work should prioritize direct experimental tests of Rotimer-mediated detection, including manipulation of web presence and structure and quantification of web geometry under native three-dimensional conditions. These approaches will be necessary to determine whether Rotimer functions as a signaling medium and to parameterize BREM for predictive application across ecological contexts.

Conclusions

These results support the interpretation that Rotimer may function as a spatially structured encounter medium, extending effective interaction distances beyond the body-contact constraints typical of planktonic systems. For benthic species at low population densities, this expansion of encounter space may be ecologically significant, a possibility that BREM makes quantitatively explicit.

By linking web structure, movement behavior, and encounter probability within a unified framework, BREM generates testable predictions and provides a foundation for comparative study across substrate-anchored biopolymer systems. Resolving the mechanistic basis of Rotimer detection and quantifying web geometry under native conditions remain key priorities, but the framework itself reframes mate search in benthic rotifers as a structured, web-mediated process rather than a passive diffusion-limited encounter.

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Data availability The datasets, ImageJ macros, R scripts, and representative video files that support this study are publicly available in the Zenodo repository at <https://doi.org/10.5281/zenodo.19153815>, enabling full reproduction of the analyses presented.

Declarations

Competing interest The authors have no relevant financial or non-financial interests to disclose.

Ethical approval No approval of research ethics committees was required to accomplish the goals of this study because experimental work was conducted with an unregulated invertebrate species (*Euchlanis dilatata*, Rotifera).

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